



Water and co-ion transport across ion-exchange membranes coated with PAH/PSS polyelectrolyte multilayer in electrodialysis and diffusion dialysis

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ABSTRACT

Ion-exchange membranes with high monovalent/divalent selectivity can accelerate the commercialization of electrochemical systems for circular economies. The membrane selectivity can be achieved by coating the membrane with a polyelectrolyte multilayer (PEM). The co-ion and water transport are critical aspects for using the coated cation-exchange membranes (CEMs) in bipolar membrane electrodialysis stacks (BMED) that utilize brine streams containing multivalent cations. In BMED, the CEM is needed to minimize the hydroxide and multivalent cation transport while facilitating the monovalent cation transport. We examined the ion and water transport across bare and coated CEMs in diffusion dialysis and electrodialysis. The CEMs were coated on one side with PAH/PSS multilayers, i.e., poly(allylamine) and poly(4-styrenesulfonate). During diffusion dialysis experiments, the ion flux across bare CEMs depends mainly on the type of anion, where $\text{OH}^- > \text{Cl}^- > \text{SO}_4^{2-}$. Moreover, the ion diffusion coefficients inside a CEM were analyzed using a Nernst-Planck model fitted to the diffusion dialysis and membrane resistance. The cation mobility is less restricted than the anion mobility inside the FKB membrane, where the trend of diffusion coefficients is $\text{Na}^+ \gg \text{Mg}^{2+}, \text{Cl}^- > \text{SO}_4^{2-}$. In PEM-coated membranes, the ion fluxes were sensitive to the coating orientation as well as the cation type. Furthermore, the PEM coating did not affect the osmotic transport of water across the coated CEMs. In other words, the water transport is limited by the permeance of the thick CEM rather than the thin PAH/PSS multilayer coating. Lastly, the effect of solution concentration on ion transport across PEM-coated membranes was examined considering the Donnan exclusion mechanism.

1. Introduction

Ion-exchange membranes (IEMs) are key components for several electrochemical technologies in water and energy applications, e.g., electrodialysis [1–4], membrane electrolysis [5], and flow batteries [6, 7]. The membrane is characterized by a high concentration of fixed-charged groups, either cationic or anionic. The function of an ion-exchange membrane is to promote counter-ion transport while restricting the water and co-ion leakages. Out of the common anions, hydroxides transport across cation-exchange membranes is of special interest. Depending on the application, the CEM is either designed to selectively facilitate the OH^- transport [8] or to minimize the OH^- leakage [9,10]. The next generation of ion-exchange membranes is required to support additional functionalities. An interesting application

for a multifunctional membrane is bipolar membrane electrodialysis (BMED). The BMED process produces acids and bases from a brine stream using electricity [11]. Standard BMED utilizes three types of membranes: a bipolar membrane, an anion-exchange membrane, and a cation-exchange membrane (CEM). The CEM separates the base stream from the salt/brine stream. The Mg^{2+} and Ca^{2+} ions can ruin the energy and current efficiency in BMED due to their high scaling potential through the formation of $\text{Mg}(\text{OH})_2$ and CaCO_3 precipitates. Hence, caustic is commonly dosed as pretreatment for the brine stream to precipitate the Mg^{2+} and Ca^{2+} ions before they reach the BMED unit [12, 13]. A CEM that can restrict both the hydroxide transport as well as the Mg^{2+} and Ca^{2+} transport is of high interest. Such a membrane will facilitate the direct use of brine in BMED without losing expensive NaOH in the brine pretreatment step.

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Polyelectrolyte multilayer (PEM) modifications can bring additional functionality to ion-exchange membranes: monovalent selective transport [14–16] and antifouling characteristics [17–19]. The PEM structure is formed by depositing layers of polycations and polyanions, alternately, via the layer-by-layer technique [20,21]. In electrodialysis, the $\text{Na}^+/\text{Ca}^{2+}$ flux selectivity across a CEM was enhanced from 0.64 (bare CEM) to 1.3 (PEM-coated CEM) [15]. Moreover, the $\text{Na}^+/\text{Mg}^{2+}$ selectivity was improved from 0.5 (bare CEM) to 1.7 (PEM-coated CEM) [21]. The monovalent/divalent selectivity can be further enhanced by tuning the PEM coating conditions as well as by selecting the optimal CEM to be coated with PEM. Coating a CEM that has a low water volume fraction leads to a more loose PEM structure, and consequently, a lower PEM resistance [16]. Under applied electric current, the PEM imposes different resistances to ions depending on the ion charge and hydration shell. Generally, the resistance of PEM-coated membranes increases in the following order of cations: $\text{K}^+ < \text{Na}^+ \ll \text{Ca}^{2+} < \text{Mg}^{2+}$ [15,16,20–22].

In our previous work [16], we analyzed the counter-ion transport across CEMs coated on one side with PAH/PSS multilayers. Interestingly, the PEM coating restricts the transport of multivalent cations, e.g., Mg^{2+} , only when the coating faces the Mg^{2+} transport direction, i.e., the forward transport direction. In the forward transport direction, the coated membrane exhibits low limiting current densities, below 4 mA/cm^2 in single electrolyte solutions of 0.25 M Mg^{2+} or Ca^{2+} . To efficiently integrate the PEM-coated CEMs in BMED stacks, the PEM coating should be oriented towards the salt compartment to restrict the Mg^{2+} and Ca^{2+} transport.

The co-ion (hydroxide) leakage across the CEMs, among other factors, can reduce the current efficiency of the BMED stack [23,24]. Moreover, the water transport across the CEMs and AEMs in BMED stacks is undesired [25–28]. On one hand, when the ion concentrations in the salt stream are much higher than the base stream, water is transported via osmosis from the base to the salt stream, which dilutes the brine stream, and resists the transport of counter-ions (cations) as the counter-ion transport via migration is opposed by the convection forces. Consequently, the water transport results in a higher energy consumption in BMED stacks. On the other hand, the water transport in the opposite direction, i.e., from the salt to the base stream, dilutes the product stream (base).

In this study, we investigate the water and co-ion transport across PEM-coated cation-exchange membranes with a focus on the coating orientation. Furthermore, we explain the interplay between the CEM properties, e.g., ionic charge density, and performance parameters, e.g., the water and hydroxide permeances. This study guides the membrane selection and design for alkali-proof membranes in BMED and alkali-selective membranes in base recovery applications.

2. Materials & methods

The investigated membranes are: Fumasep FKB-PK-130, FKS-PET-130, and FKD-PK-75 (Fumatech BWT GmbH, Germany), Selemin CMTE and CMVN (Asahi Glass Co., Japan), and Fujifilm CEM type-10 and type-12 (Fujifilm Manufacturing Europe BV, The Netherlands). The membrane properties, e.g., membrane thickness, ionic charge density, and resistance, were characterized in our previous work [29]. Solutions were prepared using Milli-Q water (Millipore) and the chemical of interest. The following reagent grade chemicals were purchased from Sigma-Aldrich: Na_2SO_4 , MgCl_2 , NaOH , poly(allylamine hydrochloride) (PAH, ~50 kDa), and poly(4-styrenesulfonic acid) solution (PSS, ~75 kDa, 18 wt% in H_2O , 1.11 g/mL at 25 °C). Sodium chloride was purchased from VWR Chemicals.

2.1. Layer-by-layer coating

We coated one side of different commercial cation-exchange membranes (CEMs) with poly(allylamine) (PAH)/poly (4-styrenesulfonate) (PSS) polyelectrolyte multilayer, as detailed in our previous study [16].

Each membrane was held horizontally using rubber gaskets and frames made of acrylonitrile butadiene styrene material (Ultimaker ABS). Firstly, a polycation layer was deposited on the cation-exchange membrane by pouring a solution of 1.9 g/L PAH (0.02 M PAH monomer unit) in 1 M NaCl, pH 2.3, on the top membrane surface. For each coating step, the polyelectrolyte solutions were left on the membrane for 5 min. After each coating step, the membrane surface was rinsed thoroughly with Milli-Q for 1 min to remove weakly attached polyelectrolytes. Afterward, a polyanion layer was deposited using a solution of 4.1 g/L PSS (0.02 M PSS monomer unit) in 0.5 M NaCl, pH 2.3. A total of 6.5 bilayers (i.e., 13 layers) were deposited on one side of the CEMs.

2.2. Diffusion experiments

Diffusion experiments were conducted to characterize the ion and water permeability across different membranes. Bare and PEM-coated samples of FKB membrane were investigated in single electrolyte solutions of NaCl, MgCl_2 , Na_2SO_4 , or NaOH. As the high solution pH can destabilize the PEM coating [30], the membrane samples used in NaOH experiments were not reused in other experiments afterward. We examined the influence of the coating orientations with respect to the ion transport direction. Furthermore, we analyzed the NaOH and water permeability across different bare CEMs: CMVN, CMTE, FKB, Fuji-CEM-10, and Fuji-CEM-12. The dialysis experiments were performed in triplicate.

Before the diffusion experiment, the membrane samples were equilibrated in the concentrated solution of interest: 0.5 M NaCl, 0.25 M MgCl_2 , 0.25 M Na_2SO_4 , and 0.5 M NaOH. To avoid having excess salt inside the membranes, the membrane samples were then submerged in the corresponding 1000x diluted solution, e.g., 0.5 mM NaCl. The membranes were then assembled into an H-cell configuration (Fig. 1a) where the membrane separates two compartments: the concentrate (e.g., 0.5 M NaCl) and the diluate (e.g., 0.5 mM NaCl). A rubber ring was placed between the membrane and the side neck of the bottle to limit water losses. The active membrane area was 8.55 cm^2 . Known masses of the solution of interest (c.a., 130 g) were poured into the respective compartments. Each compartment was capped. Moreover, a Teflon tape was wrapped around the cell joints to minimize water losses. The experiments were conducted at ambient lab temperature, i.e., 19–22 °C, for 2–3 days. In both compartments, the solutions were stirred at 150 rpm to reduce the diffusion boundary layer thickness next to the membrane surface. Over the experiment duration, samples of known volume were withdrawn from the diluate in order to analyze the ion concentration. Except for the NaOH experiments, the anion concentrations in the diluate were analyzed using ion chromatography (Dionex). For NaOH experiments, the Na^+ concentration in the diluate was analyzed using inductively coupled plasma-optical emission spectrometry (PerkinElmer ICP-OES). At the end of the experiment, the solution mass in each compartment was measured gravimetrically. The average ion and water flux over the experiment are calculated as detailed in the supporting information, S1. The measured water flux in the diffusion dialysis experiments, $J_{w,\text{net}}$ [$\text{m}^3/(\text{m}^2\cdot\text{s})$], is a combination of the osmotic and electro-osmotic water fluxes

$$J_{w,\text{net}} = J_{w,o} + J_{w,e} \quad \text{Eq. 1}$$

Under osmosis, the water moves from the diluate to the concentrate. The osmotic water flux, $J_{w,o}$ [$\text{m}^3/(\text{m}^2\cdot\text{s})$], is driven by the osmotic pressure difference generated by the ion concentration difference between the diluate and concentrate. The relation between the osmotic water flux and the osmotic pressure difference can be simplified with the water permeance, A_w [$\text{m}/(\text{s}\cdot\text{Pa})$], as a proportionality constant [31–33].

$$J_{w,o} = A_w \cdot \Delta\pi \quad \text{Eq. 2}$$

The osmotic pressure difference between the concentrate and the diluate, $\Delta\pi$ [Pa], is evaluated as follows [34,35]

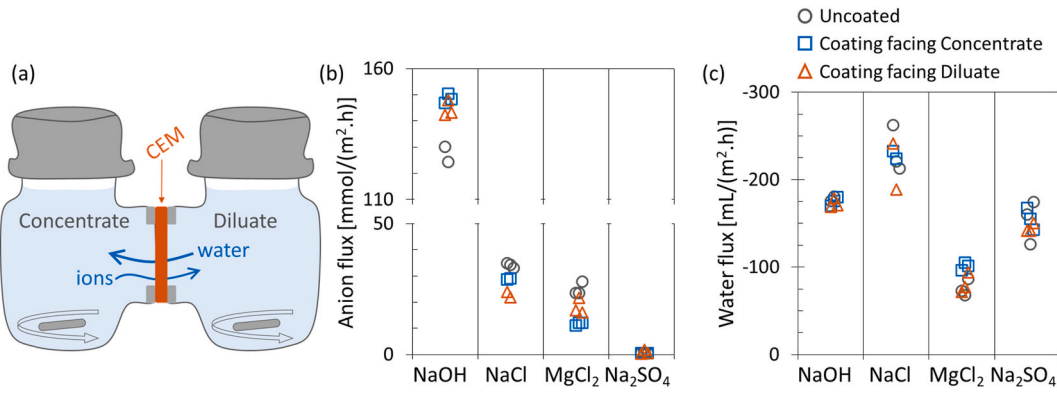


Fig. 1. (a) An illustration of the diffusion experiment. The ions diffuse from the concentrate to dilute while the net water transport is in the opposite direction. Average (b) anion and (c) water fluxes over the FKB membrane during a single electrolyte experiment of NaOH, NaCl, MgCl₂, or Na₂SO₄. The diffusion experiments were performed for three cases: bare cation-exchange membranes (circles), as well as one-side coated membranes with the coating facing either the concentrate (squares) or the dilute (triangles). The coating is composed of 6.5 PAH/PSS bilayers. The negative values of the water fluxes reflect the opposite direction of the water transport with respect to the direction of ion transport.

$$\Delta\Pi = R \cdot T \cdot \sum_i (C_i^{\text{conc}} - C_i^{\text{dilute}}) \quad \text{Eq. 3}$$

where R [J/(K·mol)] is the universal gas constant, T [K] is the temperature, and C_i^{conc} [mol/m³] is the concentration of ion (i) in the concentrate. In addition to osmosis, water molecules move via electro-osmosis while being associated with ion transport either as part of the ion hydration shell or being dragged by the ion [36,37]. The electro-osmotic flux across IEMs is quantified as follows [38]

$$J_{w,e} = \bar{V}_w \cdot \sum_i J_i \cdot n_{w,i} \quad \text{Eq. 4}$$

where J_i [mol/(m²·s)] is the ion flux, $n_{w,i}$ [mol H₂O/mol ion] is the hydration number of ion (i), $\bar{V}_w$ [m³/mol H₂O] is the molar volume of water, i.e., 18 mL/mol H₂O [38]. The electro-osmotic flux is in the same direction as the ion flux, i.e., from the concentrate to the dilute. We set the hydration numbers of Na⁺, Mg²⁺, Cl⁻, and SO₄²⁻ to 6, 15.5, 8, and 13 mol H₂O/mol ion, respectively, based on the study of Han et al. [38], where the hydration numbers were fitted to the experimental water flux during electrodialysis. Assuming a proportionality between the ion hydration number and the Stokes radius, the OH⁻ hydration number is estimated as follows

$$n_{w,\text{OH}^-} = \frac{n_{w,\text{Cl}^-} \cdot \Gamma_{\text{OH}^-}^{\text{St}}}{\Gamma_{\text{Cl}^-}^{\text{St}}} \quad \text{Eq. 5}$$

The Stokes radius of the ion, Γ_i^{St} [m], is calculated as follows [39,40]

$$\Gamma_i^{\text{St}} = \frac{\kappa_B \cdot T}{6\pi \cdot \mu_w \cdot D_i} \quad \text{Eq. 6}$$

where κ_B (1.38E-23 J/K) is Boltzmann's constant, T [K] is the temperature, μ_w (8.90E-4 Pa.s) is the water viscosity, and D_i [m²/s] is the ion diffusion coefficient. The diffusion coefficients of chlorides and hydroxides in solution at infinite dilution are 2.03E-9 and 5.27E-9 m²/s, respectively [41]. The hydration number of hydroxides is estimated as 3.1 mol H₂O/mol OH⁻.

2.3. Cyclic voltammetry and chronopotentiometry

We investigated the effect of solution concentration on the current-voltage response of different coated membranes using a 6-compartment electrodialysis cell (6C-ED, further details are provided in Refs. [16,29]). We used cyclic voltammetry to analyze the current-voltage response of coated and bare membranes in the single electrolyte solutions of NaCl, Na₂SO₄, or MgCl₂, where the cation concentration was either 0.2 or 0.5

M equivalent. Furthermore, the transient membrane response was investigated via chronopotentiometry. The steady-state current-voltage values based on chronopotentiometry are compared to the cyclic voltammograms in the Supporting Information, S3.

Before each measurement, the membrane samples were equilibrated in the solution of interest overnight. Afterward, the membrane was assembled between the middle compartments of the 6C-ED cell. The active membrane area was 23.2 ± 0.1 cm² without an area reducer, and 4.52 ± 0.04 cm² with an area reducer. The solutions were recirculated at 270 ± 10 mL/min. The solution temperature was kept at 20 ± 0.5 °C using heat exchangers. The cyclic voltammetry curves were scanned with a step of 10 mV at a rate of 5 or 0.5 mV/s using a potentiostat (Ivium Technologies, The Netherlands). The effect of solution flow rate was investigated for FKB-6.5 BL in 0.1 M MgCl₂ by measuring the membrane response at solution flow rates of 150, 270, and 390 mL/min.

3. Results & discussion

3.1. Transport in diffusion dialysis

In this section, we analyze the transport parameters for cations and anions across bare and coated cation-exchange membranes using a diffusion dialysis setup (Fig. 1a). At the start of the diffusion experiment, the cation concentration in the concentrate was 0.5 M equivalent, while the dilute was 1000x less. Ions diffused towards the dilute following their concentration gradient, while the water flowed towards the concentrate. We tested the diffusion of four different single electrolytes (Fig. 1b). For the same dialysis duration and starting concentration difference, the average OH⁻ flux was 3.7x higher than the Cl⁻ flux (Fig. 1b), which can be explained by the difference in ion mobilities and hydration numbers. In aqueous solutions, the OH⁻ exhibits 2.6x higher mobility than the Cl⁻ [41]. The ion mobility inside the membrane is reduced by a factor compared to its mobility in bulk solution, depending on the ion hydration radius and membrane water volume fraction [42]. As chlorides have larger hydrated radii compared to hydroxides [43], the chloride mobility inside the membrane is further reduced compared to the hydroxide mobility. Common ions, e.g., Cl⁻ and Na⁺, move in the solution with a hydration shell, which limits the ion mobility [34]. Instead, protons can move via the hopping/Grotthuss shuttling mechanism, where the proton is transported from a hydronium ion to a neighboring water molecule [44]. Traditionally, it was assumed that hydroxides move via hopping, similar to protons, where the hydroxide accepts a proton from one of the neighboring water molecules [34]. Recently, studies debated the transport mechanisms of hydroxides and the significance of the hopping versus the Brownian/vehicular

mechanism [45–50]. Hydroxides have lower mobility than protons, as the vehicular transport mechanism is believed to be more significant for hydroxide compared to proton transport.

We coated the CEM on one side with a PAH/PSS polyelectrolyte multilayer (PEM). The PAH/PSS multilayer is unstable at high pH [30], where the PAH loses its ionization [51]. In this regard, the PEM coating was expected to mildly reduce or not affect the NaOH diffusion as the membrane will be exposed to high pHs in the NaOH experiments. Based on the dialysis experiment, the NaOH flux was 1.2 times higher in case of the PEM-coated membrane compared to the uncoated membrane whether the coating is facing the concentrate or the diluate. Therefore, the membrane was more permeable for the NaOH transport after it was coated with the PAH/PSS multilayers. The increase in the membrane permeability after the coating can be a result of exposing the membrane to a low pH during the coating procedure (pH 2.3), followed by a high pH during the NaOH dialysis experiment (pH 10.7–13.7).

For bare-membrane experiments, the average anion flux in case of NaCl was 1.4x higher than that of MgCl₂, while it was 61x higher than that of Na₂SO₄ (Fig. 1b). Evidently, the salt diffusion across ion-exchange membranes is more limited by the co-ion type than the type of counter-ion based on their corresponding concentration inside the membrane. For NaCl and MgCl₂ experiments, the PEM coating reduced the salt flux compared to the uncoated membranes since the PEM coating imposes additional resistance against the ion transport (Fig. 1b). Moreover, the coating orientation led to different effects for each salt. The PAH is a weak polycation, while the PSS is a strong polyanion [51]. The presence of excess polycationic or polyanionic charges in the PEM can be compensated with mobile ions, i.e., extrinsic compensation [52]. The applied ionic strength and pH in coating solutions develop a PEM structure with an effective positive charge [16]. The bottleneck for NaCl diffusion across the PEM-coated membrane is the Cl[−] diffusion across the CEM. The net positive charge of the PEM layer increased the Cl[−] concentration at the PEM/CEM interface, which resulted in higher NaCl flux when the PEM is facing the concentrate relative to the opposite orientation.

Compared to NaCl, the coating orientation affected the MgCl₂ transport differently (Fig. 1b). There are two limitations on the MgCl₂ transport across the PEM-coated membranes in the diffusion dialysis. Firstly, the Cl[−] transport is limited across the CEM due to the low co-ion concentration, i.e., Cl[−], inside the CEM. Secondly, the Mg²⁺ transport is limited by the PEM coating, especially when the PEM is facing the transport direction of Mg²⁺ ions, as it leads to a lower Mg²⁺ concentration at the PEM/CEM interface. The PEM imposes a relatively high resistance against Mg²⁺ mobility compared to that of Na⁺ because Mg²⁺ ions have a larger hydration shell [16]. Moreover, the Mg²⁺ concentration in the PEM coating is lower than that of Na⁺ based on the positive charge of the PEM and the Donnan exclusion mechanism. In the Na₂SO₄ dialysis experiment, the influence of the PEM coating was negligible as

the SO₄^{2−} transport was highly restricted by the cation-exchange membrane.

In the diffusion dialysis experiments, the net water flow was in the opposite direction with respect to the ion fluxes (Fig. 1c). The PEM coating influence on the water transport was insignificant. In other words, the water transport is limited by the dense cation-exchange membrane, while the PEM coating does not add a significant resistance to water transport. This is different from the PEM-based nanofiltration membranes, where ultrafiltration membranes are coated with a PEM until the base membrane pores are fully covered [53]. Unlike PEM-coated CEMs, the water transport across PEM-coated ultrafiltration membranes is mainly limited by the PEM domain [54,55].

The osmotic water flux across a membrane is proportional to the water permeance (A_w) and the osmotic pressure difference. We analyzed the water permeance of the membrane at each electrolyte by solving Eq. (1) – Eq. (4). The membrane water permeance varied in range of $[2.2\text{--}3.5] \times 10^{-9}$ m/(s.bar) in case of Na⁺ as counter-ion, while it was $[1.2\text{--}1.5] \times 10^{-9}$ m/(s.bar) in case of Mg²⁺ (Fig. 2c). Compared to Na⁺, Mg²⁺ ions exhibit a larger hydration shell, which can reduce the free water fraction inside the membrane [29,38]. Moreover, the low mobility of Mg²⁺ inside the membrane is hypothesized to retard the water transport across the membrane, similar to the effect of Mg²⁺ on the surrounding faster cations inside CEMs [42].

In our previous work [42], we calculated counter-ion diffusion coefficients inside different cation-exchange membranes based on direct current resistance measurements. We assumed an ideal Donnan equilibrium condition where all ion activity coefficients were set to 1. In the present study, we compile the resistance measurement (performed in Ref. [42]) with the diffusion dialysis data (performed in this study) to further analyze the co-ion transport at NaCl, MgCl₂, and Na₂SO₄ electrolytes. In this regard, we evaluate the ion diffusion coefficients and activity coefficients inside the membrane by jointly fitting two sets of experiments: the diffusion dialysis and resistance measurements. The average ion fluxes of the diffusion experiment, J_i [mol/(m².s)], are modeled via the linearized Nernst-Planck equation [34] as follows

$$J_i = -D_i^m \frac{\Delta C_i^m}{\delta_m} - \frac{z_i F C_i^{m,avg} D_i^m}{RT} \frac{\Delta \phi^m}{\delta_m} + v_w C_i^{m,avg} \quad \text{Eq. 7}$$

where D_i^m [m²/s] is the ion diffusion coefficient inside the membrane, ΔC_i^m [mol/m³] is the ion concentration difference across the membrane, z_i is the ion valency, δ_m [m] is the membrane thickness, F [C/mol] is Faraday's constant, $C_i^{m,avg}$ [mol/m³] is the linear average ion concentration inside the membrane, R [J/(K.mol)] is the universal gas constant, T [K] is the temperature, $\Delta \phi^m$ is the electric potential difference across the membrane, and v_w is the water velocity inside the membrane. The fitting approach is detailed in the supporting information, S2.

In Fig. 2a, the measured concentrations from the diffusion dialysis experiment are compared to 1D transport model results, where we

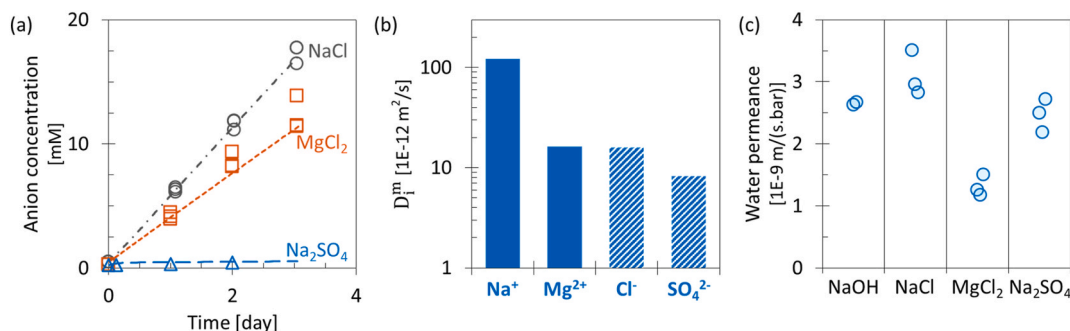


Fig. 2. (a) Anion concentration in the diluate during the diffusion dialysis experiments with a single electrolyte of NaCl, MgCl₂, or Na₂SO₄. The starting anion concentration was 0.5 mM and 0.5 M equivalent in the diluate and concentrate, respectively. The diluate and concentrate were separated by a standard cation-exchange membrane, i.e., FKB. The markers represent the experimental results, while the dashed lines refer to the model results. (b) Fitting values for the ion diffusion coefficients inside FKB membranes. (c) Water permeance of an FKB membrane in a single electrolyte of NaCl, MgCl₂, or Na₂SO₄.

utilized the fitted diffusion and activity coefficients as model inputs. The model is described in our previous work [56]. The fitted activity coefficients and diffusion coefficients inside FKB membrane are reported in Table 1 and Fig. 2b. The Na^+ diffusion coefficient in the membrane is an order of magnitude higher than that of Mg^{2+} . Compared to the ion mobility in aqueous solution, the Na^+ inside the membrane is reduced by a factor of 11, while the Mg^{2+} mobility is reduced by a factor of 43 (Table S2). The ion mobility reduction in the membrane is caused by the membrane tortuosity [29] as well as the ion hydration volume and energy [42]. The co-ion mobility is reduced by a factor of c.a. 130 for Cl^- and SO_4^{2-} . As a counter-ion leaves its position, another counter-ion moves to replace it [42,57]. As the co-ion concentration is much lower than that of counter-ions, co-ions are less likely to exchange positions with a surrounding ion. Instead, the co-ion transport is expected to incur more friction with the membrane polymer.

The ion activity coefficients inside the membrane can be used to calculate the ion concentrations via the Donnan equilibrium condition (Eq. S10). At ideal conditions (at infinite dilution), the ion activity coefficients equal 1. Our fitting resulted in a co-ion activity coefficient of 1.8 for the FKB membrane in MgCl_2 solution (Table 1). A higher co-ion activity coefficient inside the membrane translates to a lower co-ion concentration inside the membrane. Therefore, the co-ion concentration in the membrane based on an activity coefficient of 1.8 is lower than the predictions assuming ideality (activity coefficient of 1). Possibly, the low co-ion concentration (relative to the ideal predictions) is a result of the limited space inside the cation-exchange membrane of FKB, especially in the presence of counter-ions of a relatively large hydration shell, such as Mg^{2+} ions.

3.2. Influence of ion concentration on PEM resistance

The PAH/PSS polyelectrolyte multilayer coatings impose high resistance towards the transport of multivalent cations, e.g., Mg^{2+} and Ca^{2+} , with a negligible increase in the resistance towards monovalent cations, e.g., Na^+ and K^+ [16]. In a 0.1 M Mg^{2+} solution, the current-voltage response of the PEM-coated CEM exhibited an asymmetric behavior with respect to the current direction (Fig. 3a). When the PEM coating is facing the Mg^{2+} transport direction (forward transport, Fig. S2b), the current-voltage curve reaches a limiting current density as the Mg^{2+} ions are depleted in the PEM at the PEM/CEM interface. Increasing the solution flow rate reduces the diffusion boundary layer at the membrane surface. However, increasing the solution flow rate did not influence the limiting current density of the PEM-coated CEM in 0.1 M MgCl_2 (Fig. 3a). Within the current range of $\pm 4 \text{ mA/cm}^2$, the current-voltage response of the bare FKB membranes is linear in 0.1–0.25 M MgCl_2 solutions (Fig. 3b). Hence, the bare membranes were operated in the under-limiting current regime, and there are sufficient ions in the diffusion boundary layers to conduct the current. The limiting current behavior of the PEM-coated membranes in MgCl_2 solutions is a result of the ion depletion inside the PEM coating, and it does not depend on the diffusion boundary layers.

The coated membrane response can be described by two parameters: the area resistance and the limiting current density. In MgCl_2 solution, the limiting current density of the PEM-coated membrane increases as the solution concentration increases (Fig. 3b). For FKB coated with 6.5 PAH/PSS bilayers, increasing the Mg^{2+} concentration in solution by

2.5x (from 0.1 to 0.25 M) leads to a 2.3x increase in limiting current density (from 0.25 to 0.58 mA/cm^2 , Fig. S5a).

In bipolar membrane electrodialysis (BMED), high current densities, in the range of 20–100 mA/cm^2 , are typically applied to operate the bipolar membrane in the over-limiting current regime [12,58,59]. To evaluate the potential application of the PEM-coated membranes in BMED stacks, we scanned the cyclic voltammograms of the bare and coated membranes in 0.2 and 0.5 M NaCl solutions in a wide range of currents, i.e., ± 40 and $\pm 80 \text{ mA/cm}^2$, respectively (Fig. 3c). In NaCl solutions, the current-voltage response of the coated membrane is similar to that of the bare membranes. Unlike the transport of multivalent cations, the PAH/PSS coatings do not impose significant resistance towards Na^+ transport. The current-voltage curves of the coated and uncoated FKB membranes exhibit a limiting current regime at high applied current densities due to the ion depletion in the diffusion boundary layer. In a spacer-free channel with a linear solution velocity of c.a. 0.4 cm/s, the limiting current density of the bare FKB membrane was $25 \pm 1 \text{ mA/cm}^2$ in 0.2 M NaCl (Fig. S5b). The applied current density in BMED should be high enough to operate the bipolar membrane in the overlimiting current regime. At the same time, applied current density should be below the limiting current density of the PEM-coated CEMs, which depends on the ion concentrations in the feed stream of the BMED stack, the solution flow rate, and the spacer hydrodynamics [60–62].

The total area resistance of the coated membrane, $\Lambda_{\text{m+PEM}}$ [Ωm^2], is calculated as follows [16]

$$\Lambda_{\text{m+PEM}} = A_{\text{m}} \cdot (r_{\text{m+PEM+s}} - r_{\text{s}}) \quad \text{Eq. 8}$$

where A_{m} [m^2] is the active membrane area, $r_{\text{m+PEM+s}}$ [Ω] is the combined resistance of the coated membrane and solution, and r_{s} [Ω] is the resistance measured in the stack with the solution but without the membrane in place. In the present study, the membrane resistances are calculated based on direct current measurements, which account for the membrane resistance as well as the interfacial resistances at the membrane surface, i.e., the electrical double layer and the diffusion boundary layer [29,63–65]. In MgCl_2 solutions, the combined resistance, $r_{\text{m+PEM+s}}$, is evaluated based on the slope of the current-voltage data points in the reverse direction, i.e., $[0, -4 \text{ mA/cm}^2]$, as explained in Ref. [16]. In NaCl solutions, the combined resistance is evaluated based on the ohmic range of currents, i.e., $[-10, 10 \text{ mA/cm}^2]$, in order to reduce the effect of the ion depletion in the diffusion boundary layer on the membrane resistance calculations.

In MgCl_2 solutions, increasing the solution concentration reduces the coated membrane resistance, especially for low resistance membranes, e.g., FKB (Fig. 3d). The influence of the solution concentration on the LCD and membrane area resistance of the coated membranes is a consequence of the ion partitioning between the solution and the PEM domain. The PEM domain has a net positive charge, which limits the presence of the cations according to the Donnan exclusion mechanism. The multivalent cations are further excluded compared to the monovalent cations. Furthermore, the ions with larger hydration radius and energy, e.g., Mg^{2+} , have lower mobility compared to ions with smaller hydration shells, e.g., Na^+ .

The coated membrane resistance increases by about an order of magnitude when the cation type is changed from Na^+ to Mg^{2+} (Fig. 3e). The PAH/PSS coating is selective for the transport of the monovalent cations, e.g., Na^+ , over the multivalent ones, e.g., Mg^{2+} , based on the Donnan exclusion mechanism and the difference in the ion mobilities. Regarding the influence of the anion type, the coated membrane resistance did not change significantly in response to changing the anion type from Cl^- to SO_4^{2-} . Moreover, the current-voltage response of the coated membrane was linear and symmetric with respect to the current direction in 0.5 M Na^+ , whether the anion is Cl^- or SO_4^{2-} (Fig. 3c, Fig. S6). Therefore, the anion charge, i.e., -1 or -2 , does not affect the cation transport across PEM-coated CEMs in electrodialysis application. That is

Table 1

Fitted values for the ion activity coefficient inside the membrane at single solution electrolytes of NaCl, MgCl_2 , or Na_2SO_4 .

Solution electrolyte	γ_i^{m}	
	cation	anion
NaCl	1.0	1.0
MgCl_2	1.0	1.8
Na_2SO_4	1.0	1.4

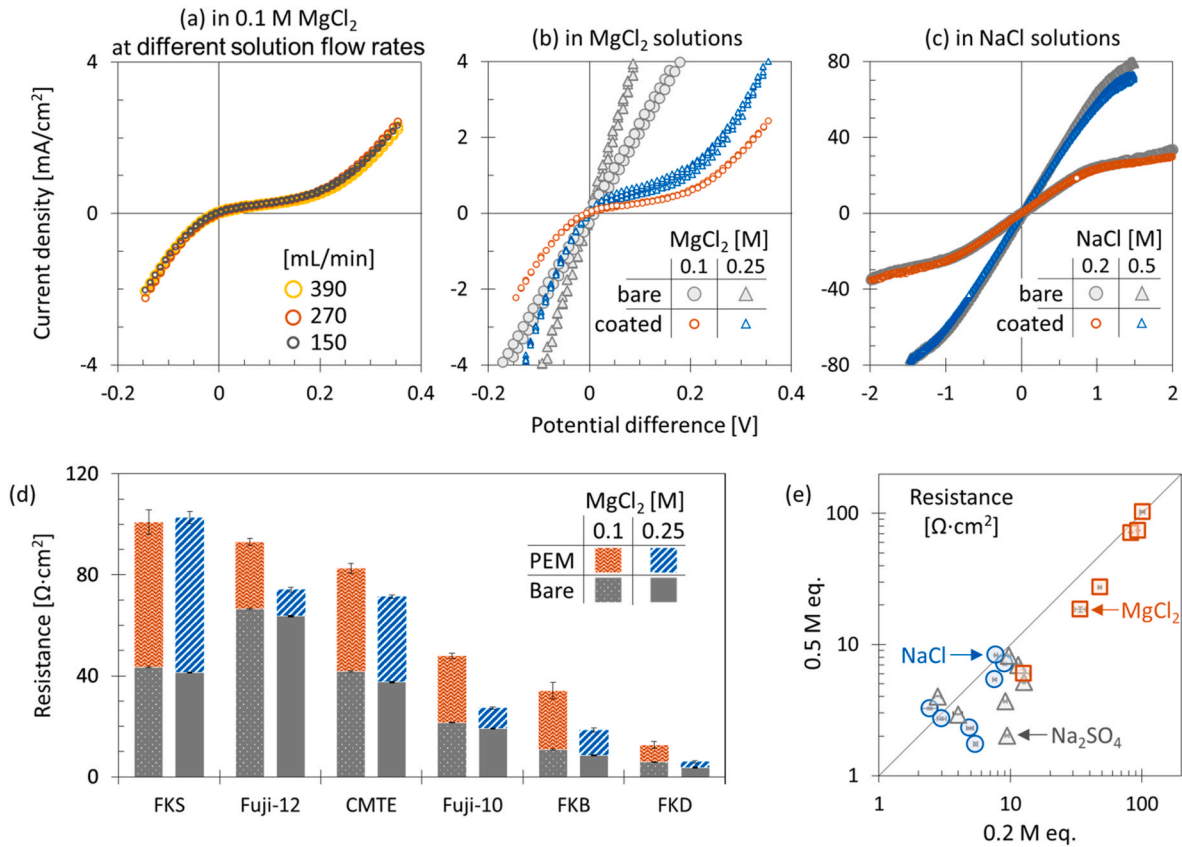


Fig. 3. (a) The current-voltage response of the coated membrane (FKB with 6.5 PAH/PSS bilayers) in 0.1 M MgCl_2 solution at different solution flow rates: 150, 270, and 390 mL/min. Influence of (b) MgCl_2 and (c) NaCl concentration in solution on the current-voltage response of bare and coated FKB membranes. The membrane was coated on one side with 6.5 PAH/PSS bilayers. The membrane response was characterized using a 6-compartment electro dialysis cell and cyclic voltammetry. All the cyclic voltammograms were scanned at 5 mV/s, except for the PEM-coated FKB membrane in MgCl_2 solutions, where the scan rate was 0.5 mV/s. (d) The resistance of six different bare (grey bars) and coated cation-exchange membranes in 0.1 and 0.25 M MgCl_2 solutions. The membranes were coated on one side with a polyelectrolyte multilayer (PEM) of 6.5 PAH/PSS bilayers. The increase in the membrane resistance after coating is illustrated with the orange and blue bars. (e) The resistance of PEM-coated cation-exchange membranes in different single electrolyte solutions of NaCl, Na_2SO_4 , or MgCl_2 . The cation concentration in the solution was 0.2 M equivalent (x-axis) or 0.5 M equivalent (y-axis). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

totally different from the diffusion dialysis applications, where the cation and anion fluxes across the CEMs strongly depend on the anion charge based on the Donnan exclusion mechanism (Fig. 1b).

3.3. Hydroxide leakage in CEMs

The co-ion leakage in CEMs is a critical aspect for hydroxide-related applications. We characterized the NaOH and water diffusion across 6 different commercial CEMs. The NaOH flux depends on the partitioning inside the membrane as well as the NaOH permeance, B_{NaOH} [m/s], as follows

$$J_{\text{NaOH}} = B_{\text{NaOH}} \cdot \Delta C_{\text{NaOH}}^m \quad \text{Eq. 9}$$

where J_{NaOH} [mol/m².s] is the average NaOH flux across the CEM, and ΔC_{NaOH}^m [mol/m³] is the NaOH concentration difference between the two membrane sides. The co-ion partitioning mainly depends on the membrane ionic charge density, C_{fix}^m . For the NaOH experiment, the OH^- concentrations inside the membrane are evaluated via the ideal Donnan equilibrium and the electroneutrality condition as follows [29]

$$C_{\text{Na}^+}^m \cdot C_{\text{OH}^-}^m = C_{\text{Na}^+} \cdot C_{\text{OH}^-} \quad \text{Eq. 10}$$

$$C_{\text{Na}^+}^m - C_{\text{OH}^-}^m - C_{\text{fix}}^m = 0 \quad \text{Eq. 11}$$

The membrane resistance, as well as the NaOH and water permeances, are connected to the membrane water volume fraction via the

membrane tortuosity and the membrane-ion or membrane-water friction [42]. The membrane resistance in NaCl is a commonly measured parameter for CEMs. In this regard, correlating the membrane NaOH permeance to the membrane resistance in NaCl is useful to benchmark the commercial or novel CEMs. Based on characterized membranes in this study, the NaOH and water permeances are highly correlated to the membrane resistance via power law fittings (Fig. 4a)

$$A_w = 8.0159 \cdot \Lambda_{m,\text{NaCl}}^{-1.276}, \text{ with } R_D^2 = 0.96 \quad \text{Eq. 12}$$

$$B_{\text{NaOH}} = 6.6831 \cdot \Lambda_{m,\text{NaCl}}^{-1.074}, \text{ with } R_D^2 = 0.94 \quad \text{Eq. 13}$$

where A_w [1E-9 m/(s.bar)] is the water permeance of the membrane in NaOH electrolyte, $\Lambda_{m,\text{NaCl}}$ [$\Omega \cdot \text{cm}^2$] is the membrane area resistance in 0.5 M NaCl as measured via the direct current method, B_{NaOH} [1E-6 m/s] is the NaOH permeance, and R_D^2 is the coefficient of determination. The permeance is an extensive property of the membrane that depends on the membrane thickness. The permeability is an intensive property of the material, as it is normalized for the membrane thickness as follows

$$\beta_w = A_w \cdot \delta_m \quad \text{Eq. 14}$$

$$\beta_{\text{NaOH}} = B_{\text{NaOH}} \cdot \delta_m \quad \text{Eq. 15}$$

where β_w and β_{NaOH} are the water and NaOH permeabilities, respectively. The water and NaOH permeabilities are linearly correlated to the

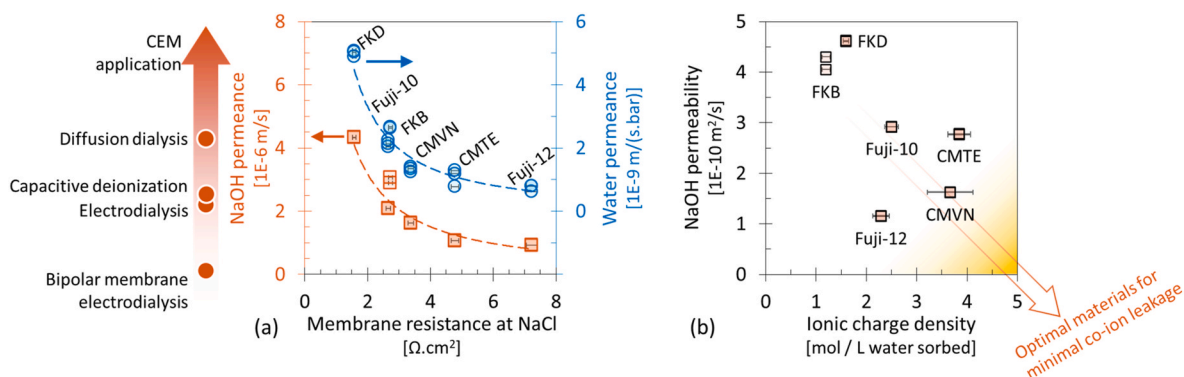


Fig. 4. (a) The NaOH (left y-axis) and water (right y-axis) permeances through different cation-exchange membranes versus the membrane resistance. The NaOH and water permeances were analyzed based on diffusion experiments. The membrane resistance was measured using a 6-compartment electro dialysis at 0.5 M NaCl via potential sweep. The dashed lines represent power law fittings. (b) The NaOH permeability through different cation-exchange membranes versus the membrane ionic charge density. The ICD values are retrieved from Refs. [29,42].

Na^+ diffusion coefficient as follows (Fig. S7)

$$\beta_w = 0.0416 \cdot D_{\text{Na}^+}^m - 0.3711, \text{ with } R_D^2 = 0.94 \quad \text{Eq. 16}$$

$$\beta_{\text{NaOH}} = 0.035 \cdot D_{\text{Na}^+}^m + 0.1314, \text{ with } R_D^2 = 1.00 \quad \text{Eq. 17}$$

where $D_{\text{Na}^+}^m$ [$1\text{E-}12 \text{ m}^2/\text{s}$] is the Na^+ diffusion coefficient inside the membrane, β_w [$1\text{E-}13 \text{ m}^2/(\text{s} \cdot \text{bar})$] and β_{NaOH} [$1\text{E-}10 \text{ m}^2/\text{s}$] are the water and NaOH permeabilities across a CEM, respectively. The Na^+ diffusion coefficients inside the membrane are calculated based on the measured membrane resistance (direct current method) at 0.5 M NaCl according to Ref. [42]. The water content, ionic charge density (ICD), and thickness of an IEM can be tuned to achieve a specific co-ion flux based on the application requirements. The higher the membrane ICD is, the lower the co-ion partitioning inside the membrane becomes (Eq. (10), Eq. (11)). Moreover, the lower the membrane water content is, the lower the ion diffusion coefficients and the NaOH permeability are (Fig. S7). Hence, a membrane with a high ICD and a low water volume fraction is ideal to minimize co-ion leakage (Fig. 4b). However, there is a trade-off between the membrane swelling and ICD. Typically, membranes of higher ionic charge densities swell more [66–69]. In principle, the membrane thickness can be tuned to match the required permeance. Practically, the choice of membrane thickness is limited by further criteria, e.g., the mechanical stability and the aging behavior of the membrane [70,71].

3.4. Implications for BMED

Bipolar membrane electro dialysis (BMED) processes utilize electricity and a pretreated brine stream to produce acid and base [11, 72–74]. To maintain a high current efficiency, the BMED stack includes a CEM to retain the produced OH^- ions and reduce water crossover. An extensive pretreatment of the brine is currently required to remove the Mg^{2+} and Ca^{2+} ions, as they limit the BMED efficiency by forming scales when combined with hydroxides and carbonates, respectively [13,75]. This pretreatment is costly as it consumes a substantial amount of NaOH. To boost the economics and sustainability of this BMED application, the next generation of CEM should include additional functionality by preventing the transport of multivalent ions from the brine stream toward the base stream. This functionality can enable the direct usage of untreated brine in the BMED stack and minimize the cost of brine pretreatment. In this regard, coating a polyelectrolyte multilayer (PEM) on one side of the CEM is of high interest. The CEM hinders the OH^- transport from one side, while the PEM coating hinders the Mg^{2+} and Ca^{2+} transport from the other side. Fuji-CEM-12 had the lowest NaOH permeance ($9\text{E-}7 \text{ m/s}$) among the investigated membranes (Fig. 4a). In terms of the isotropic properties of the membrane, the CMVN material

has a relatively low OH^- permeability and high ICD (Fig. 4b), which is ideal for OH^- blocking.

The second requirement on the CEMs is to hinder the transport of multivalent cations, e.g., Mg^{2+} , which can be achieved by coating the CEM with PAH/PSS polyelectrolyte multilayer (Fig. 3d). The choice of the CEM that will be coated with PEM depends on the ICD and water volume fraction of the bare membrane [16]. A high resistance towards Mg^{2+} transport can be achieved by coating a CEM that has a low water volume fraction and a moderate ICD. In this regard, FKS and Fuji-CEM-12 are good candidates to be coated with the PEM to block the transport of Mg^{2+} ions. For future work, Fuji-CEM-12 coated with 6.5 PAH/PSS bilayers is expected to provide a good balance to block the OH^- and Mg^{2+} transport in the BMED application.

4. Conclusion

We characterized the ion and water transport across coated and bare cation-exchange membranes (CEMs). The diffusion of salts across bare CEMs is mainly determined by the anion type, e.g., Cl^- , SO_4^{2-} , or OH^- , as well as the membrane properties, including the membrane water content and ionic charge density. The ionic charge density determines the anion partitioning inside the CEM. Furthermore, the co-ion permeability across the membrane increases as the membrane water volume fraction increases. We coated different commercial cation-exchange membranes on one side with a polyelectrolyte multilayer (PEM). For uncoated CEMs, the water permeance correlates well with the ionic resistance of the membrane, as both parameters depend on the membrane water content. Coating the CEM with PAH/PSS multilayer did not affect the osmotic water transport. In this regard, the coated PEM structure is hypothesized to have a higher water permeance compared to the CEM domain. CEMs of low water content are proposed to be coated on one side with PAH/PSS multilayers and used in bipolar membrane electro dialysis (BMED) applications to minimize the transport of the hydroxides and problematic multivalent cations, such as Mg^{2+} .

CRedit authorship contribution statement

Alaaeldin A.E. Elozeiri: Writing – original draft, Visualization, Supervision, Software, Methodology, Investigation, Formal analysis. **Seline Zijderveld dos Santos:** Methodology, Investigation. **Jouke E. Dykstra:** Writing – original draft, Visualization, Supervision, Methodology, Funding acquisition, Formal analysis. **Huub H.M. Rijnaarts:** Writing – original draft, Visualization, Supervision, Methodology, Funding acquisition, Formal analysis. **Rob G.H. Lammertink:** Writing – original draft, Visualization, Supervision, Methodology, Funding acquisition, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.memsci.2025.125072>.

Data availability

The data presented in this paper is available at the 4TU Research Database via <https://doi.org/10.4121/29fca47e-8ccf-4499-b65e-ea01041b06c8>

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